Mobile IP Overview

By
Sunirmal Khatua
University of Calcutta

What is Mobile IP?

- It is an Open Standard defined by IETF that extends IP.
- Allows user to keep same IP address, stay connected & maintain ongoing applications while roaming between different IP Networks.
- Based on IP – so any media that can support IP can also support Mobile IP
- It provides Mobility Transparency over IP to higher level protocols like TCP (TCP/MobileIP).
- It allows transparent routing of IP datagram to and from Mobile devices over IP network.
- Mobile IPv4 is defined in IETF RFC 3344 &
  Mobile IPv6 is defined in IETF RFC 3775

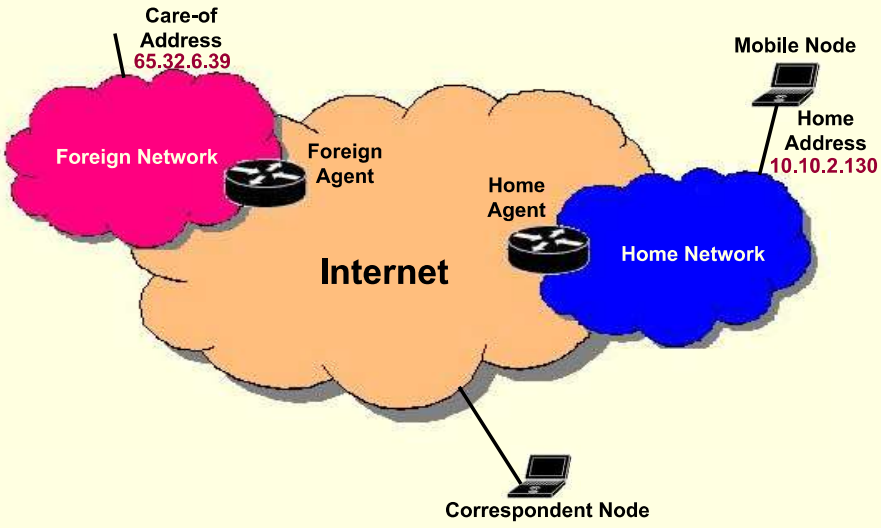
What is Mobile IP?

- In TCP/IP protocol, IP determines next hop for an IP Packet by using the Destination IP within the packet and a Routing Algorithm.
  - *Thus IP Address must be changed whenever a Device changes its point of attachment to the IP Network.*
- In TCP/IP protocol, TCP maintains connections that are indexed by
  <Source IP, Source Port, Destination IP, Destination Port>
  - *Thus IP Address must not be changed to maintain TCP connections.*
- Mobile IP is the Solution which is a Standard defined by IETF that extends IP.
  - Mobile IPv4 is defined in IETF RFC 3344 &
  - Mobile IPv6 is defined in IETF RFC 3775
- Allows user to keep same IP address, stay connected & maintain ongoing applications while roaming between different IP Networks.

How Mobile IP Works?

- Definitions:
  - **Mobile Node** – Host/Router that can change its point of attachment to the IP Network.
  - **Home Address** – Static IP Address for a Mobile Host.
  - **Home Network** – The Network, possibly virtual, whose Network ID matches with that of a Mobile Host's Home Address.
  - **Home Agent** – A router on the Mobile Host's Home Networks that is responsible maintaining the current location of a Mobile Host when it is away from the Home Network and for forwarding IP traffic to that Mobile Host
  - **Foreign Network** – A network that a Mobile Host visits when away from its Home Network.
  - **Care-of Address** – A temporary IP Address that is assigned to a Mobile Host when it visits a Foreign Network.
  - **Foreign Agent** – A router on the Mobile Host's Foreign Network that provides routing service to the Mobile Node while registered.
  - **Correspondent Node** – A peer host with which a Mobile Host is communicating. It may be mobile or stationary.

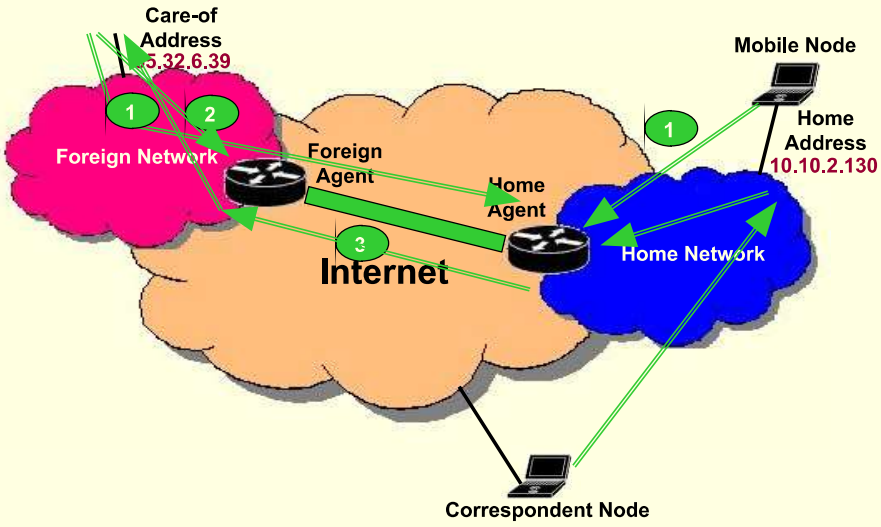
Definitions (Cont.)



Working of Mobile IP

- The Mobile IP works in three main phases:
  1. Agent Discovery:
     - A Mobile Node discovers its Home Agent & Foreign Agent
  2. Registration:
     - The Mobile Node registers its current location with Foreign Agent & Home Agent
  3. Tunneling:
     - A reciprocal tunnel is setup by the Home Agent to the Care-of Address to route packets to the Mobile Node as it roams.

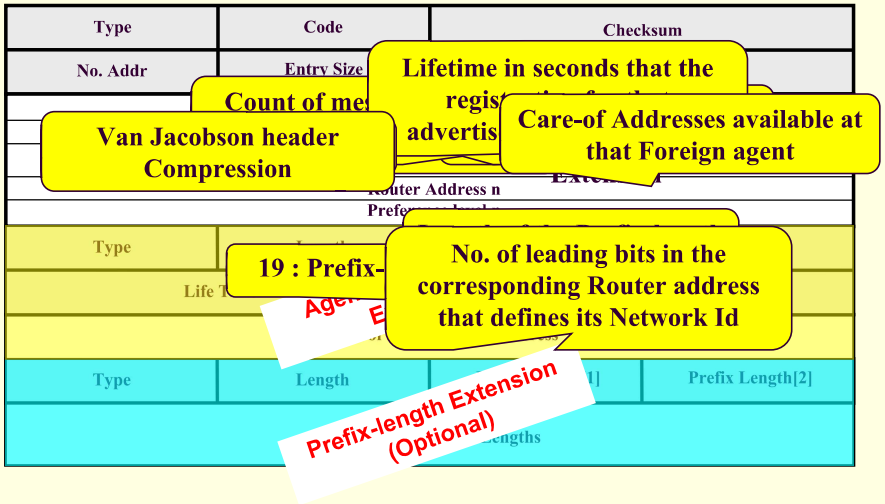
Working of Mobile IP (Cont.)



Agent Discovery

- Mobility agents advertise their presence by periodically broadcasting Agent Advertisement messages using **ICMP Router Discovery Protocol (IRDP) with advertisement extension.** The message consists of
  - IP address of the router.
  - Zero or more advertised Care-of Addresses.
  - Some additional information.
- The Mobile Node listens to these advertisements to determine if it is connected to its home network or foreign network
  - If the Network Id of the advertised IP Address match with that of the Home Address, then it is in Home Network otherwise in Foreign Network.
- If a mobile node does not wish to wait for the periodic advertisement, it can send out Agent Solicitation messages that will be responded by a mobility agent.

IRDP Message Format with Extension



Agent Discovery (Cont.)

- Move Detection:
  - The mobile node may move from one network to another without the IP level being aware of it. The Agent Discovery process enable the agent to detect such a move.
  - The agent use one of the following two algorithms for the purpose.
    1. **Use of Lifetime Field:**
       - ✔ After receiving the message the node uses the Lifetime field as a Timer. If the timer expires before the next advertisement, the node assumes that it has lost contact with the agent.
       - ✔ If, in the meantime, the node receives an advertisement from another agent then it registers with the new agent.
       - ✔ Otherwise it uses Agent Solicitation to find an Agent
    2. **Use of Network Prefix:**
       - ✔ It the Network Id of a newly received Advertisement doesn't match with the current Care-of address then it assumes that it have moved to another network and registered accordingly.

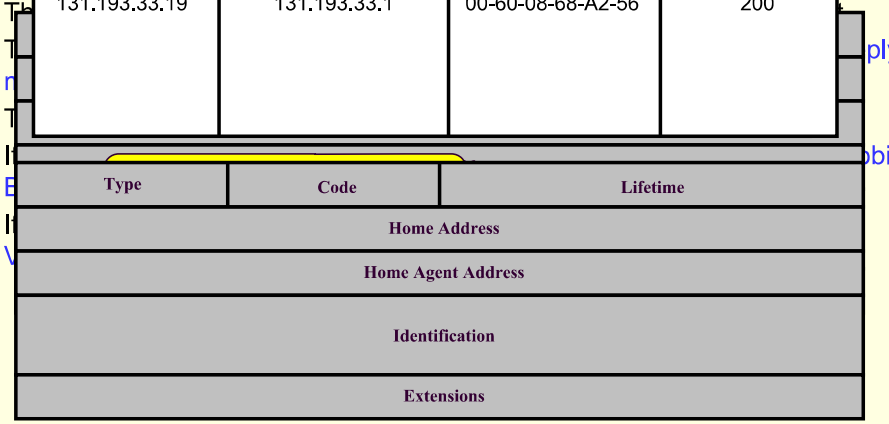
Agent Discovery (Cont.)

- Types of Care-of Address:
  1. Foreign agent Provided Care-of Address:
     - The Care-of address is an IP address for the Foreign agent
     - Thus the Foreign Agent will receive any packet intended for the Mobile Host and will forward it to that Host
  2. Co-Located Care-of Address:
     - Co-Located Care-of Address is required if there are no Foreign Agent in the Network or all the Foreign Agents are busy.
     - Co-Located Care-of Address can be obtained by DHCP or by an IP address owned by the Mobile Host for Foreign Networks.

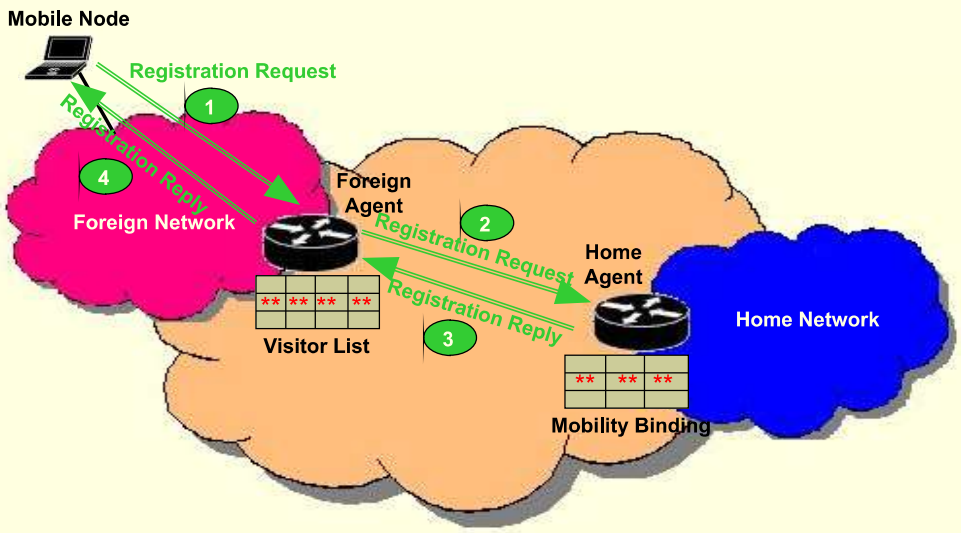
Registration

- If a mobile node discovers that it is on the Home Network, it operates without an
- If t with the
- Th
- T ply n
- T
- It bility E
- It V

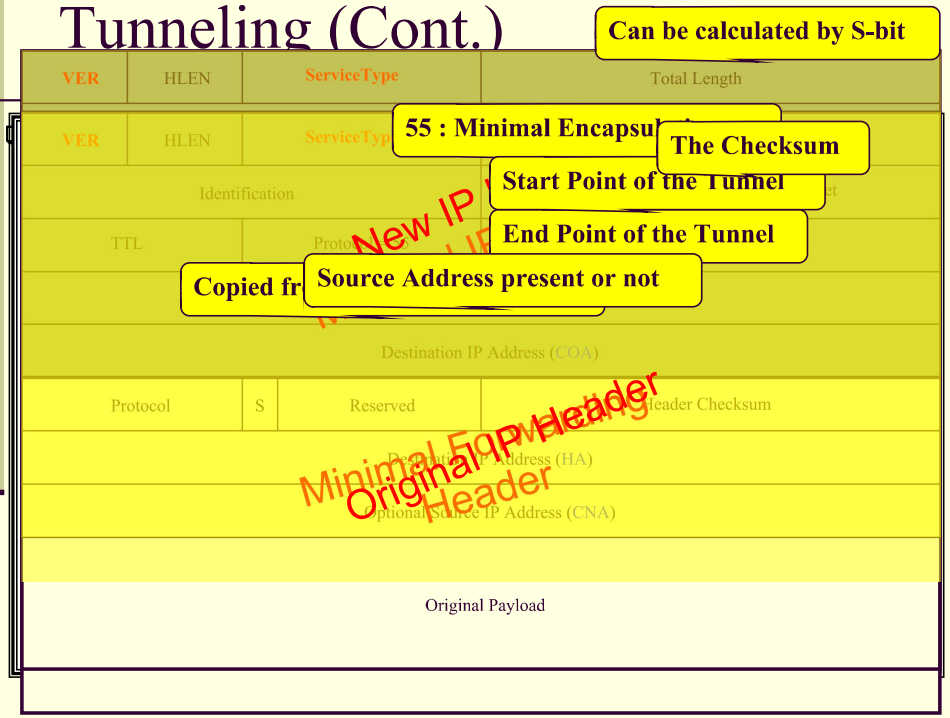
Home Address	Home Agent Address	MAC Address	Lifetime (sec.)
131.193.44.14	131.193.44.7	00-60-08-95-66-E1	150
131.193.33.19	131.193.33.1	00-60-08-68-A2-56	200



Registration (Cont.)



Tunneling

- In Mobile IP, the Correspondent Node sends IP datagram to the Home Address. The IP datagram is captured by the Home Agent.
  - Whenever a Mobile Node choses a Home Agent, the agent inform all other nodes in that network that any datagram destined to that Mobile Node should be delivered to that agent.
- The Home Agent checks its Mobility Binding Table to find the Care-of Address of the Node if it is visiting a Foreign Network.
- The Home Agent encapsulates the IP datagram within another IP destined to the Care-of Address.
  - If the Node uses Foreign Agent provided Care-of Address then the destination will be the Foreign Agent.
  - If the Node uses Co-located Care-of Address then the destination will be the Node itself.
- The receiver (Foreign Agent or the Node) decapsulates the datagram and forward the original IP datagram to the Node.
- This process of encapsulation and decapsulation of a datagram is frequently referred to as “Tunneling"

Tunneling (Cont.)